

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

## Chapter 2: Polynomials

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the zeroes of  $x^2 + 9x + 14$  and verify the relationship between the zeroes and the coefficients.
2. Find the zeroes of  $9x^2 - 12x + 4$ .
3. Find the zeroes of  $5x^2 - 20$ .
4. Find a quadratic polynomial whose zeroes have sum 4 and product  $-5$ .
5. Find a quadratic polynomial whose zeroes are  $\frac{2}{3}$  and  $-\frac{1}{2}$ .
6. If  $\alpha$  and  $\beta$  are the zeroes of  $x^2 - 6x + 8$ , find  $\alpha^2 + \beta^2$ .
7. If  $\alpha$  and  $\beta$  are the zeroes of  $3x^2 + 5x - 2$ , find  $\frac{1}{\alpha} + \frac{1}{\beta}$ .
8. If one zero of  $x^2 - 8x + k$  is 6, find  $k$  and the other zero.
9. If the sum of the zeroes of  $kx^2 + 2x + 3k$  is equal to their product, find  $k$ .
10. The zeroes of  $x^3 - 4x^2 + x + 6$  are  $-1, 2$  and  $3$ . Verify the three relations between the zeroes and the coefficients.

## Answer Key

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1.  $-2$  and  $-7$ ; sum  $= -9$ , product  $= 14$

2.  $\frac{2}{3}$  and  $\frac{2}{3}$

3.  $2$  and  $-2$

4.  $x^2 - 4x - 5$

5.  $6x^2 - x - 2$

6.  $36 - 16 = 20$

7.  $\frac{-5/3}{-2/3} = \frac{5}{2}$

8.  $k = 12$ ; the other zero is  $2$

9.  $-\frac{2}{k} = 3$ , so  $k = -\frac{2}{3}$

10. Sum  $= 4$ ; sum of products in pairs  $= -2 + 6 - 3 = 1$ ; product  $= -6$